$$S_0 \xrightarrow{15} S_1 \xrightarrow{8} S_2 \xrightarrow{12} S_3 \xrightarrow{6} S_4 \xrightarrow{4} S_5$$

$V(S_0) = 40 \quad V(S_1) = 30 \quad V(S_2) = 20 \quad V(S_3) = 10 \quad V(S_4) = 5 \quad V(S_5) = 0$

MONTE CARLO $\qquad V(s_t) \leftarrow V(s_t) + \alpha\left[G_t - V(s_t)\right]$

$$V(s_4) = 5 + \frac{1}{2}\left[\underbrace{4 - 5}_{-1}\right] = 4.5$$

$$V(s_3) = 10 + \frac{1}{2}\left[\underbrace{10 - 10}_{0}\right] = 10$$

$$V(s_2) = 20 + \frac{1}{2}\left[\underbrace{22 - 20}_{2}\right] = 21$$

$$V(s_1) = 30 + \frac{1}{2}\left[\underbrace{30 - 30}_{0}\right] = 30$$

$$V(s_0) = 40 + \frac{1}{2}\left[\underbrace{45 - 40}_{5}\right] = 42.5$$

TD 0 $\qquad V(s_t) \leftarrow V(s_t) + \alpha\left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\right]$

$$V(s_0) = 40 + \frac{1}{2}\left[\underbrace{15 + 30 - 40}_{5}\right] = 42.5$$

$$V(s_1) = 30 + \frac{1}{2}\left[\underbrace{8 + 20 - 30}_{-2}\right] = 29$$

$$V(s_2) = 20 + \frac{1}{2}\left[\underbrace{12 + 10 - 20}_{2}\right] = 21$$

$$V(s_3) = 10 + \frac{1}{2}\left[\underbrace{6 + 5 - 10}_{1}\right] = 10.5$$

$$V(s_4) = 5 + \frac{1}{2}\left[\underbrace{4 + 0 - 5}_{-1}\right] = 4.5$$